

AR Smart Glasses for Home Applications – A Focus for Those with Disabilities

Statement of Problem

As society enters a new age of artificial intelligence, more home-application devices attempt to implement this life-changing technology. Companies such as Meta and Google are even trying to implement AI through augmented reality hardware and software; however, there is a major issue: a lack of artificial intelligence technology for disabled individuals. For decades, artificial intelligence has been constantly developed to cater towards the average human, yet handicapped people have struggled to utilize the global revolution of AI and instead rely on hearing aids, long canes, and screen readers. This project proposes a **solution to change the lives of the disabled** by using augmented reality (AR) smart glasses as a bridge to the AI breakthrough, using the glasses as an aid for hearing, sight, and movement navigation.

Introduction

Augmented reality smart glasses are primarily used for the average consumer, especially with the new Meta AI Glasses that target social media users. However, these **glasses do not target all audiences such as those with disabilities**, especially those who are blind or deaf and cannot connect to the AI boom. The proposed project will utilize augmented reality glasses that integrate artificial intelligence object recognition to help all audiences benefit from AI efficiently. It involves input from a camera and speaker to generate outputs, like visual and hearing aids, to support the handicapped. **The overarching problem for external human support will be solved using these glasses**, providing independence.

The project's goals are as follows:

- **Design an augmented reality system** using glasses to display visual and audio data
- Build the **electronics system** that connects a PCB, waveguide module, waveguide drivers, batteries, speakers, microphones, and a camera
- Code a **YOLO object recognition AI model** in Python to accurately detect objects in the real-world such as other humans
- Code a **Kotlin and Java application on a Linux WebOS software** to display the object recognition data to the user including audio outputs
- **Test and gather data** about the glasses' efficiency, accuracy, temperatures, and battery life to determine what improvements need to be changed on the next iteration

Pi Chart for Percentage of Disabled Individuals Who Have Access to Artificial Intelligence (2024)

As shown in the Pie Chart (Fig 1.1), there are as little as **9%** of disabled individuals who actively use AI in daily life. This number shows **no significant growth** for future impact.

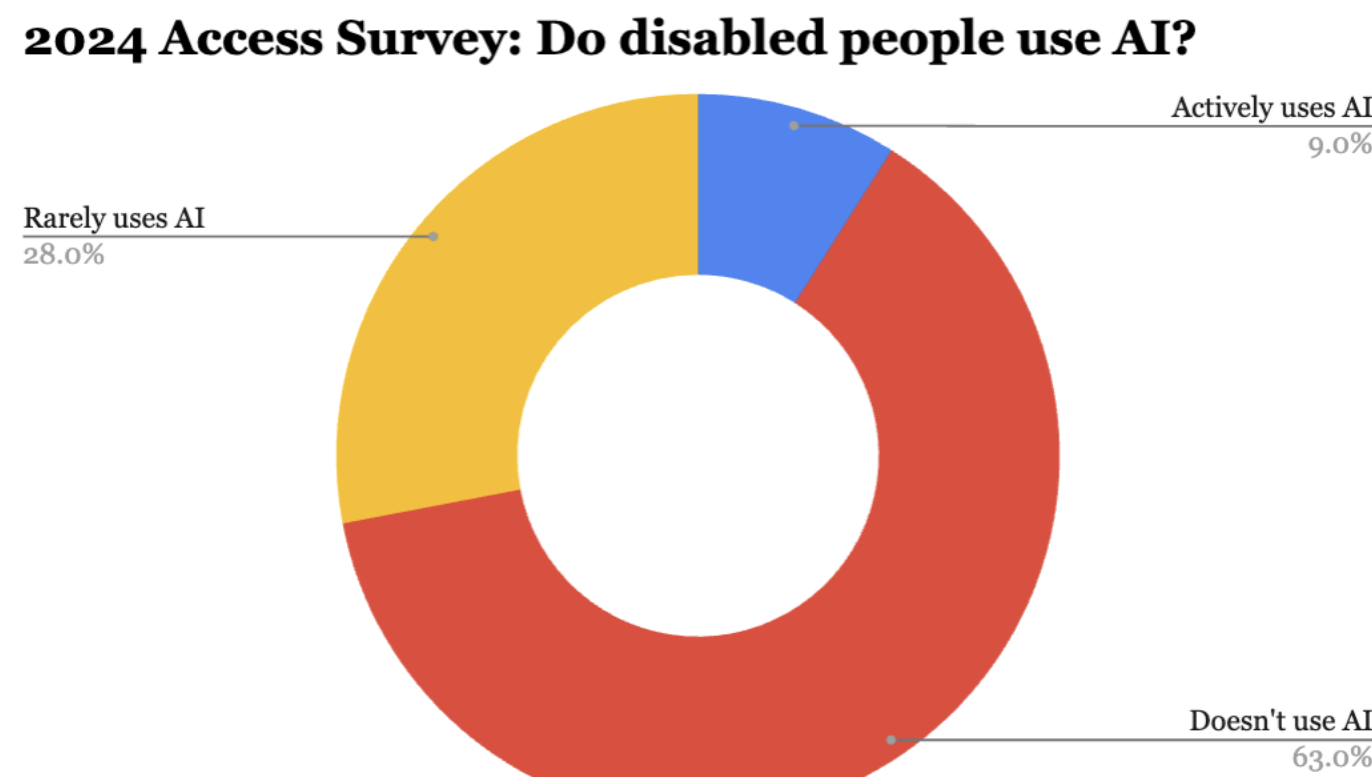


Figure 1.1. 2024 Access Survey: Do disabled people use AI?
Source: EuansGuide.com

The data distribution in the pie chart is concerning because **huge AI companies ignore the importance** for disabled individuals who need assistance.

This project focuses to change this number as it **lays out the foundation for further application usage**. Glasses, as an everyday wearable accessory, **can be used for more than its intended purpose**: to provide autonomy to those who need support from those around them.

Materials

AR Smart Glasses

- x2 G502 Waveguide Module
- x2 G502 Waveguide Driver
- x2 Waveguide Lens Almond Shape
- x1 ESP32-S3 PCB
- x1 OmniVision OV2640 Camera
- PMIC Li-ion Charger
- Rotary Controller
- Miniature Buttons
- LED Strip
- x2 Li-ion 4.2V Battery 150 mAh
- Lead-free Solder Wire
- Weller 60W Soldering Iron
- x1 Digital MEMS Microphone
- x1 Rectangular Micro Dynamic Speaker
- x1 9-axis ICM-42688 IMU Sensor
- Ribbon Cables
- Metal Shields
- 388 Neutral Cure Silicone Glue
- Scotch Transparent Tape

Computer-Aided Design (CAD)

- Autodesk Inventor CAD Software
- Bambu Labs P1S 3-D Printer
- Bambu Labs PLA Filament (Yellow)
- Bambu Studio

HUD Application

- Kotlin
- Java
- Android Studio
- Android 15.0 SDK
- Temurin-17 JDK
- Core Runtime 2.1.0
- **YOLO AI Model**
- PyCharm (Python)
- TensorFlow Library
- Ultralytics Library
- Common Objects in Context Dataset
- RTX 4070 GPU
- Google Colab
- Common Objects in Context Dataset

Methods

3-D Design

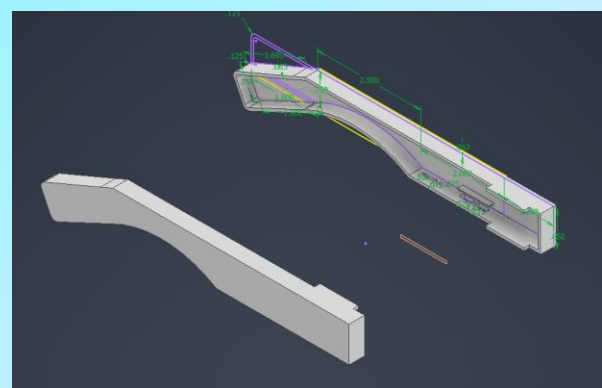


Figure 2.1. Glasses CAD in Inventor
Source: Author

The first step in the engineering process is to CAD the glasses in Inventor (Fig 2.1) where electronics and structure is designed.

Build



Figure 2.2. Interior of Left Arm
Source: Author

The glasses model was built for testing (Fig 2.2) and was ready to be programmed for real world testing environments.

Code

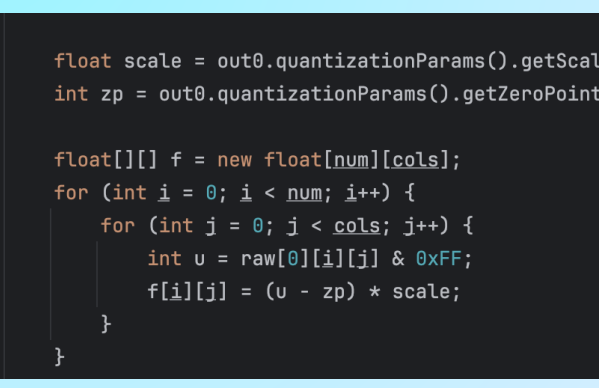


Figure 2.3. Running YOLO Model
Source: Author

Displays and the AI for the glasses were coded in in both Android Studio and Google Colab. Data was gathered and tested (Fig 2.3).

Test and Analyze

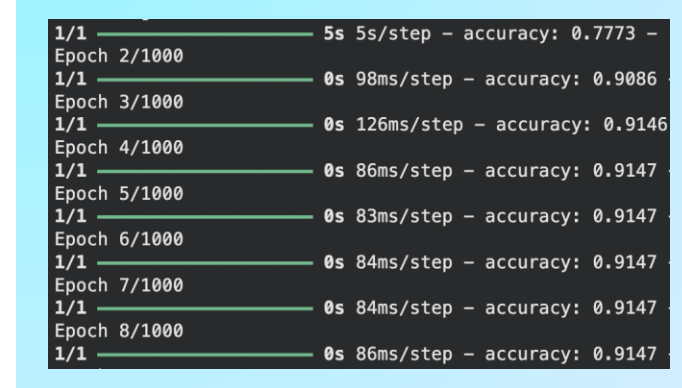


Figure 2.4. YOLO AI Accuracy Test
Source: Author

The displays and AI model were tested individually. Specifically, the AI model was trained using a 20/80 split (Fig 2.4).

3-D Printing Glasses Frame



Figure 3.1. 3-D printed frame for glasses
Source: Author

The frame of the glasses were printed using **PLA filament on a Bambu Labs P1S 3-D Printer** (Fig 3.1). Multiple iterations of the glasses' frames and arms were modeled and printed for improvement.

Gathering Data for Glasses Input

Getting precise and accurate data from the glasses is an important step in determining the reliability for the platform. To get data from the glasses, the sensors would **send their returned data through a USB-C cable** from the PCB to the terminal (Fig 5.1). This data was analyzed in the PCB and sent into an AI YOLO object recognition model that would **determine specific outputs to send back to the display HUD**. After further analysis on the accuracy of the AI model, logical and concise conclusions would be performed to ensure the prototype is ready for real-world application.

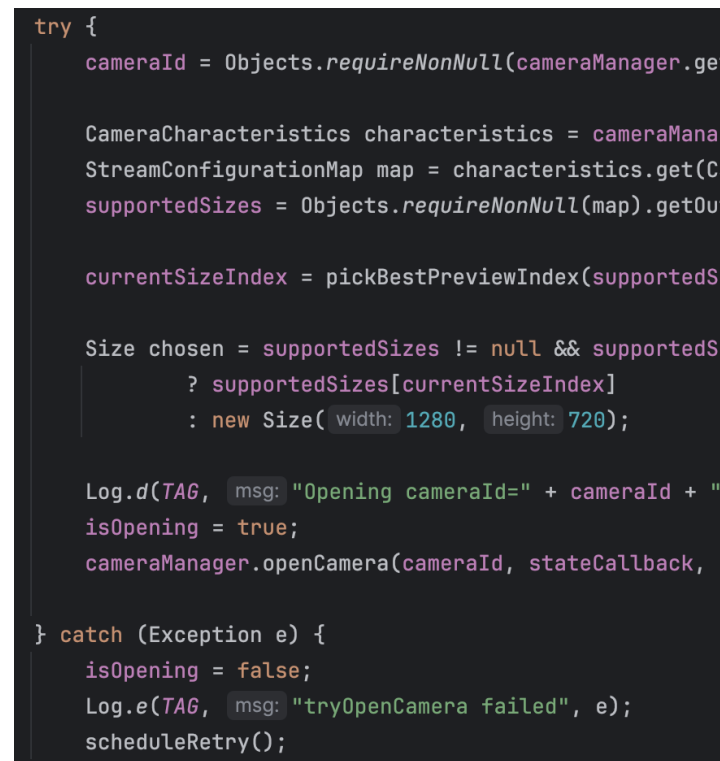


Figure 5.1. Pulling camera data in code
Source: Author

Results

Glasses Structure

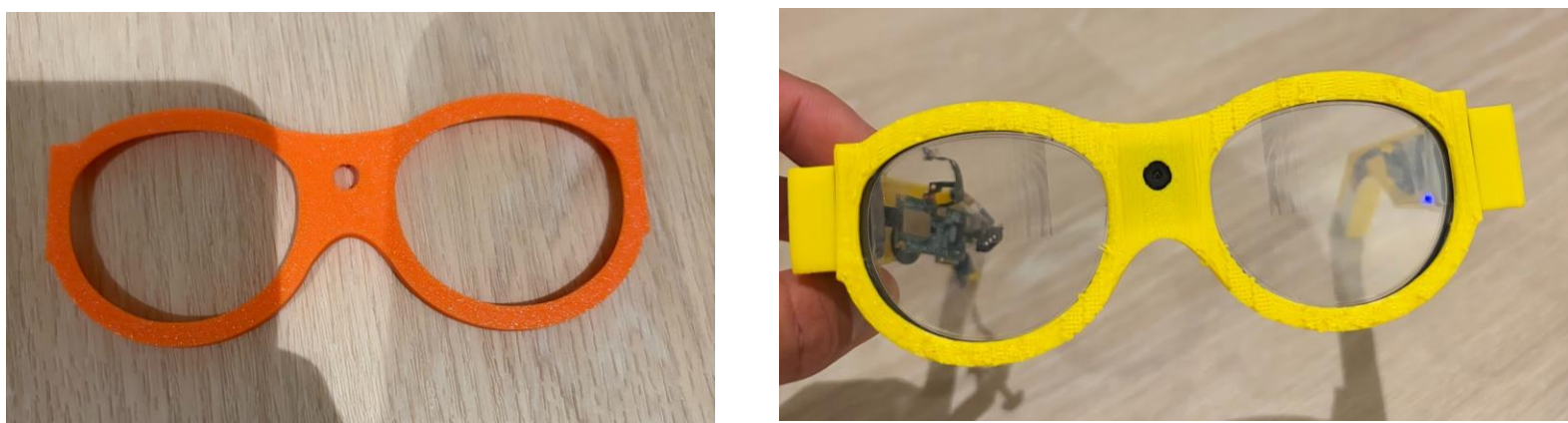


Figure 6.1. Glasses' Lens Iterations
Source: Author

The structure of the glasses is crucial for both comfort and durability. **Multiple iterations** for the frame were CADed and tested. The walls were printed at **1/16 inch thick**, creating a strong layer to protect the electronics inside. The frame contains **custom-made buttons** and a **camera hole** (Fig 6.1). **The arms are also angled down** slightly to lower the center of gravity (Fig 6.2).

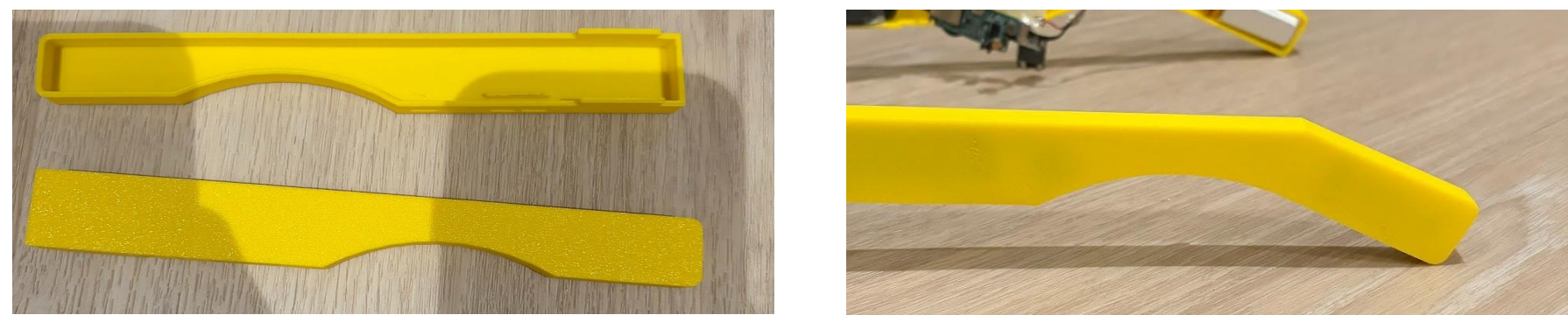


Figure 6.2. Glasses' Arms Iterations
Source: Author

Electronics Configuration

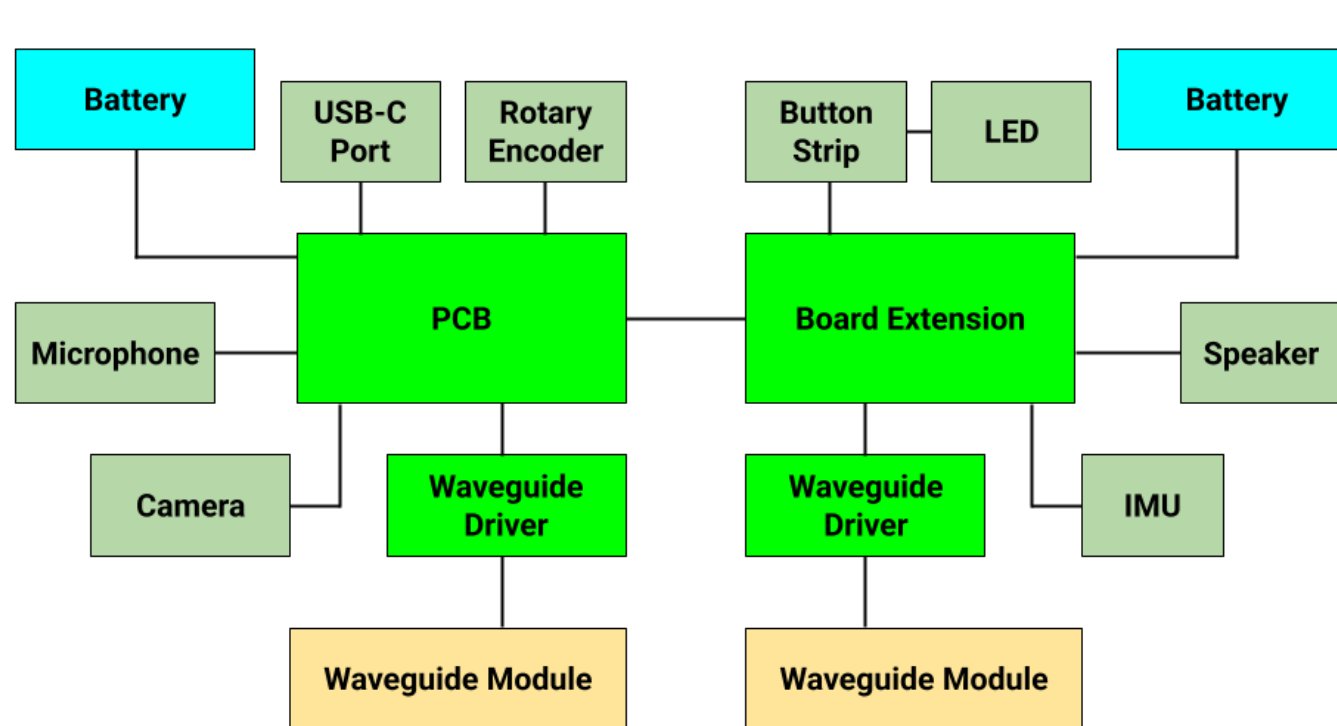


Figure 8.1. Circuit diagram of electronics
Source: Author

This circuit diagram (Fig 8.1) **visualizes how everything is connected**. The PCB is wired to a board extension to obtain more ports for connections. **Sensors are distributed evenly across both boards** and are wired directly, while the **G502 Waveguides are connected through the drivers**.

Testing Glasses Accuracy

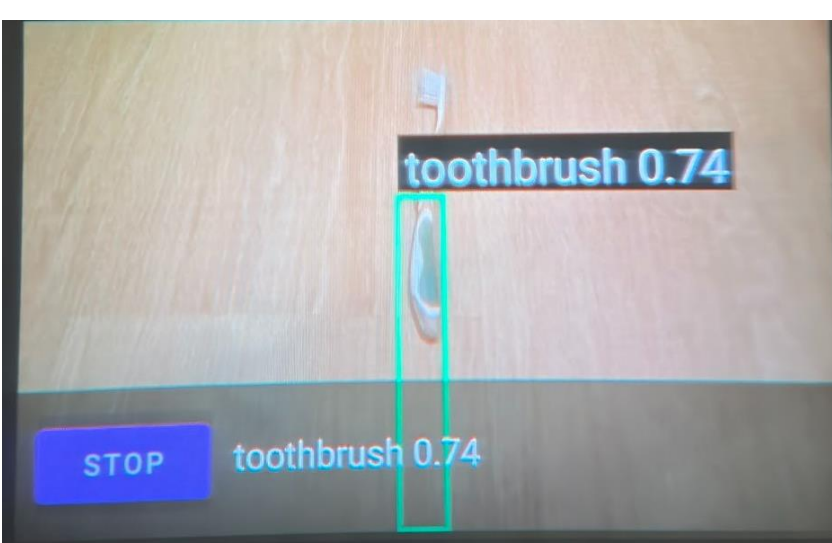


Figure 10.1. Accuracy Test on Toothbrush
Source: Author

Item	Accuracy (%)
Person	72
Toothbrush	73
Bottle	71

Table 1.1. YOLO Accuracy on Items
Source: Author

After **20 trials** for each item, the YOLO AI model was able to **sustain an accuracy of over 70% on average**. Tests were conducted on a person, a toothbrush, and a water bottle. This accuracy may seem low, but for a pair of glasses that is running an AI model on an **ESP32-S3 chip the size of a penny**, these results prove to be excellent.

YOLOv8 Image Recognition

These are the equations used for YOLO object detection (Eqn 1.1). It calculates bounding boxes, scores for each image, box error, and if a box should be shown. This AI model was trained on **300 epochs** and **168 layers** using the COCO dataset.

$$(x, y, w, h) = (\sigma(t_x) + c_x, \sigma(t_y) + c_y, p_w e^{t_w}, p_h e^{t_h})$$

$$p_{obj} = P(\text{object}) \cdot \text{IoU}$$

$$\text{Score}_i = p_{obj} \cdot P(\text{class}_i | \text{object})$$

$$\mathcal{L} = \mathcal{L}_{\text{box}} + \mathcal{L}_{\text{obj}} + \mathcal{L}_{\text{class}}$$

$$\text{Keep } b \text{ if } \text{IoU}(b, b_{\text{max}}) < \tau$$

Equations 1.1. YOLO Equations
Source: Author

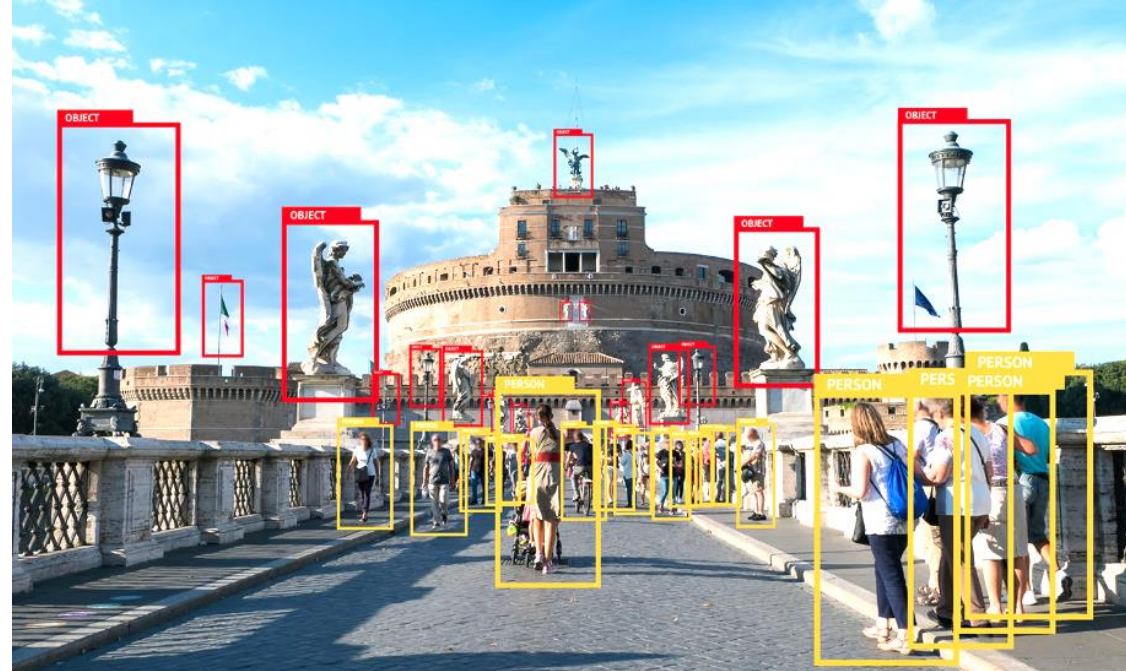


Figure 7.1. YOLO Bounding Boxes Example
Source: medium.com

AR Glasses Linux Software

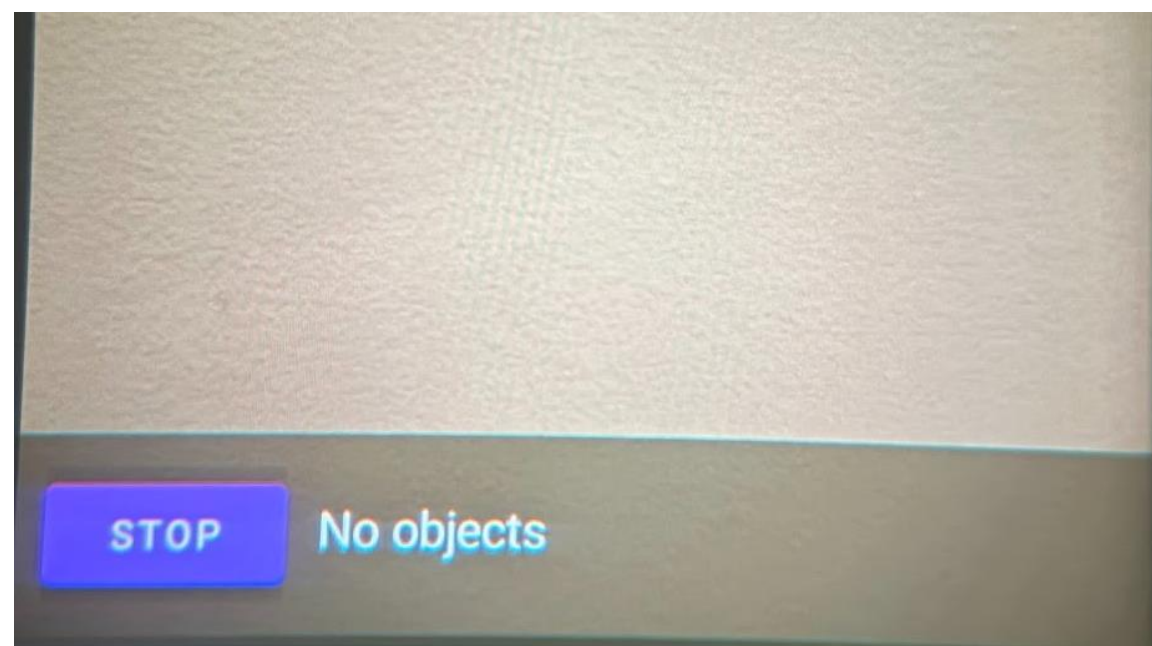


Figure 9.1. Glasses App Frontend
Source: Author

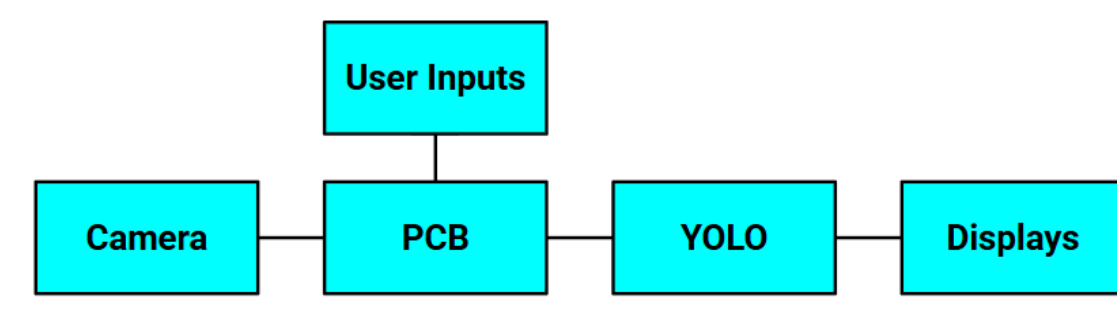


Figure 9.2. Diagram of Output Flow
Source: Author

Analyzing Data

The data from the sensors on the glasses proves to be accurate in contrast to its small size. The glasses can feed video and audio data to the PCB with **little noise from the outside environment**. In fact, the camera is amazing at capturing photos and videos in high resolution (Fig 11.1). Due to extra space in the right arm, a **microphone upgrade is a viable choice** for better sound input quality.



Figure 11.1. OmniVision OV2640 Camera
Source: Author

Discussion

The results of this study demonstrate the effectiveness of how the AR Smart Glasses for Home Applications as it successfully outlined an environment that supports those with disabilities. The data collected from over 60 trials revealed that the glasses correctly estimated certain objects with an **average success rate of 72.4%**. The glasses had a **runtime of approximately 2.128 hours** and could **detect objects within 109.3 milliseconds**, much faster than the average human reaction time. The significance of these results are highlighted in comparison to previous studies (Fig 12.1) on augmented reality systems. In the study, Ibrahim and his team tested object recognition using YOLO, trained on the COCO dataset, by using the **Microsoft HoloLens augmented reality headgear** (Fig 12.2). Surprisingly, the speed of detection was **slower than the current implementation with an average speed of approximately 200 milliseconds**.



Figure 12.1. Ibrahim Research Experiment Using YOLO
Source: Ibrahim, 2022



Figure 12.2. Microsoft First Generation HoloLens
Source: microsoft.com

The hypothesis that integrating augmented reality with AI object detection and displays was strongly supported by the collected data. Compared to expensive and bulky headwear like the HoloLens, **this prototype was able to process detections faster at a cheaper price** as the prototype has a cost around **\$600**, whereas the HoloLens is estimated to cost around **\$3,500**.

The effectiveness of the Augmented Reality Smart Glasses for Home Applications system proves to be a dominant foundation for human support, allowing those with disabilities to function more independently. By detecting objects rapidly, these glasses **negate the necessary use for walking canes for the blind**, as the YOLO model in combination with the glasses' speakers can **determine obstacles in the environment effectively**. However, every solution is not perfect and there are a few minor issues with the glasses design. First, the glasses are **front-heavy**, making the accessory fall off the face easily. Furthermore, the glasses **commonly experience overheating, going past 50 °C after 10 minutes** of continuous use of the AI model. Lastly, the glasses' arms are a bit wide and blocky, making the glasses feel a bit **uncomfortable after long hours of use**.

In conclusion, this study provides strong evidence that augmented reality accessory systems are **extremely beneficial for human support**. The rapid detection rate through the YOLO AI model explains the superiority of using AI in real-world environments. Furthermore, AR displays can also be beneficial for those who are deaf or mute. As augmented reality technology continues to advance, incorporating AI is **essential in achieving peak human efficiency and support for one another**.

Future Implications

This device for augmented reality systems is far from finished. This current prototype is only designed around those who are blind; however, these glasses have the potential to support humans with all disabilities. For example, the augmented reality display can be used to help the deaf hear others by translating conversations in real time. Furthermore, the hardware in these glasses does not have the best possible version to keep its cheap cost for consumers. In the future, this project **plans to upgrade current systems and test new methods for the glasses**:

- **Develop a new YOLO AI model** that is trained using more layers and images, increasing the reliability for object recognition. A model this complex is **currently unable to be implemented due to hardware limitations**. Therefore, after a hardware upgrade, this new AI system is entirely possible.
- **Upgrade the glasses with multiple or stronger PCBs** with the addition of **better waveguide modules**, allowing the glasses' frames per second to be faster and create a higher resolution display for quality of life.
- **Code and test new display systems** to adapt to those with other disabilities. Testing **new systems to support the deaf and mute are a priority** for this prototype as this project aims to connect to all audiences.

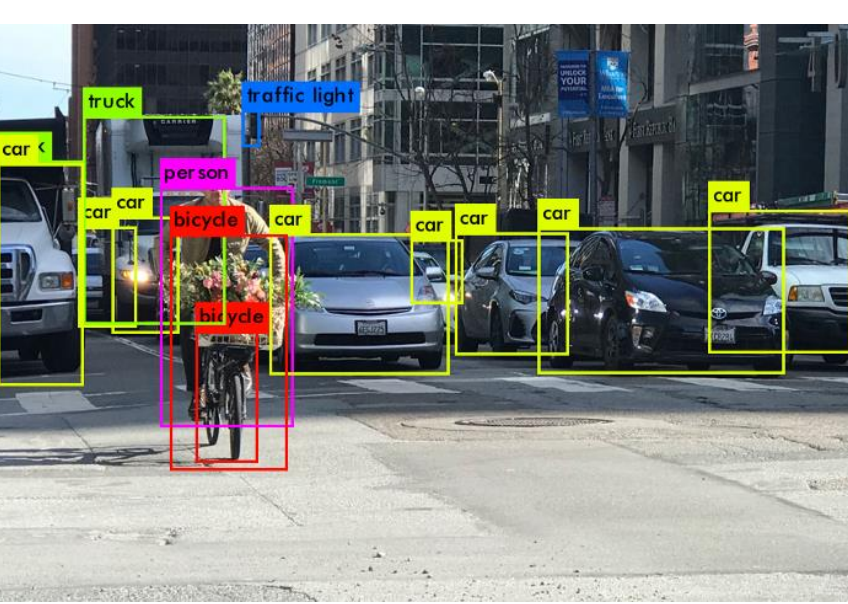


Figure 13.1. YOLO Model with More Layers
Source: mathworks.com

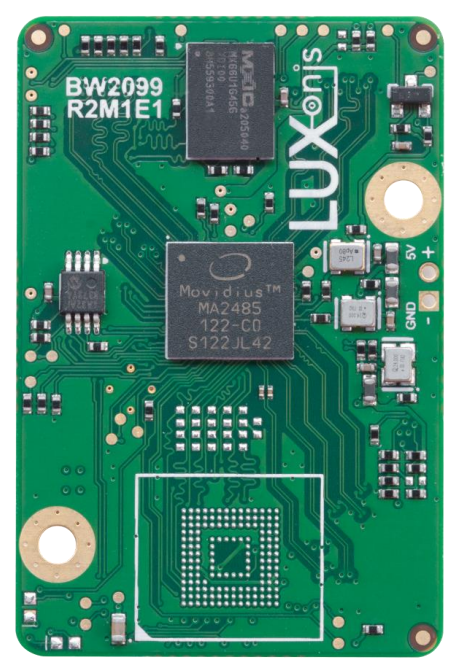


Figure 13.2. Luxonis OAK-SoM
Source: luxonis.com

This project plans to continue this extraordinary research, focusing on an implementation for more advanced techniques. As more disabled individuals are limited by AI, this technology will prove to be useful in **supporting lives daily**.